



Course Code: CSE 3200
Course Title: Design Project-2
Section: 03
Semester: Summer '26

Requirement Analysis Report

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Requirement Analysis Report

Software Requirements Specification (SRS)

Project : InnerBloom - Mental Wellbeing Management System

Date : June 20, 2026

Introduction

InnerBloom is a web-based Mental Wellbeing Management System designed to support mental healthcare services by connecting patients, doctors, and administrators on a single platform. The system aims to provide convenient access to mental health support through online appointment booking, self-assessment tools, AI chatbot assistance, and secure user management.

The purpose of this system is to improve accessibility, efficiency, and organization in mental healthcare services while maintaining user privacy and security.

Stakeholders & Users

Role	Description	Primary Goal
Patient	Individuals seeking mental health support and consultation.	Access mental health services easily and manage appointments.
Doctor	Mental health professionals providing consultation services.	Manage appointments and provide support to patients.
Administrator	System manager responsible for maintaining the platform.	Manage users, verify doctors, and monitor system activities.

Functional Requirements

These requirements define what the system should do

3.1 Authentication Module

- **FR-01:** The system shall allow patients to register using their name, email, and password.
- **FR-02:** The system shall allow doctors to apply for registration by providing professional information.
- **FR-03:** The system shall allow all users to log in using valid credentials.

3.2 Patient Management

- **FR-04:** Patients shall be able to create and update their profiles.
- **FR-05:** Patients shall be able to view available doctors and their information.
- **FR-06:** Patients shall be able to search doctors by specialization.
- **FR-07:** Patients shall be able to view appointment history.

3.3 Appointment Management

- **FR-08:** Patients shall be able to book appointments with doctors.
- **FR-09:** Doctors shall be able to view appointment requests.
- **FR-10:** The system shall store appointment records for future reference.
- **FR-11:** The system shall send appointment reminders to users.

3.4 Mental Health Self-Assessment

- **FR-12:** The system shall provide mental health self-assessment questionnaires.
- **FR-13:** The system shall calculate assessment results automatically.
- **FR-14:** Patients shall be able to view their assessment history.

3.5 AI Chatbot Module

- **FR-15:** The system shall provide an AI chatbot for basic mental wellbeing guidance.
- **FR-16:** Users shall be able to interact with the chatbot through a chat interface.
- **FR-17:** The chatbot shall provide general mental health information and support.

3.6 Doctor Management

- **FR-18:** Doctors shall be able to submit enrollment applications.
- **FR-18:** Doctors shall be able to update professional profiles.
- **FR-19:** Doctors shall be able to manage their appointment schedules.

3.7 Admin Module

- **FR-20:** Administrators shall be able to approve or reject doctor applications.
- **FR-21:** Administrators shall generate unique doctor IDs.
- **FR-22:** Administrators shall manage patient and doctor accounts.
- **FR-23:** Administrators shall monitor appointments and system activities.
- **FR-24:** Administrators shall manage subscription plans.

Non-Functional Requirements

These requirements define the quality of the system

4.1 Performance

- **NFR-01:** The system should load pages within 3 seconds under normal conditions.
- **NFR-02:** Appointment booking requests should be processed within 2 seconds.

4.2 Security

- **NFR-03:** User passwords must be stored using encryption.
- **NFR-04:** Only authorized users should access protected resources.
- **NFR-05:** User mental health information must remain confidential.

4.3 Reliability

- **NFR-06:** The system should be available 99% of the time.
- **NFR-07:** Appointment and assessment records should be stored without data loss.

4.4 Usability

- **NFR-08:** The system should have a user-friendly interface.
- **NFR-09:** Users should be able to navigate the system without special training.

Data Requirements

The system will store the following entities:

1. **Patient:** PatientID, Name, Email, Phone Number, Password, Profile Information
2. **Doctor:** DoctorID, Name, Email, Specialization, Qualification, Experience, Verification Status
3. **Appointment:** AppointmentID, PatientID, DoctorID, Appointment Date, Appointment Time, Status
4. **Assessment:** AssessmentID, PatientID, Assessment Date, Score, Result
5. **Chatbot Interaction:** ChatID, PatientID, Message, Response, Timestamp
6. **Subscription:** SubscriptionID, PatientID, Plan Type, Start Date, End Date

Assumptions & Constraints

- **Constraints**
 - The system will be developed as a web application.
 - The backend database will use MySQL.
 - Stable internet access is required for chatbot and appointment services.
 - The system will not provide emergency mental health services.

- **Assumptions**

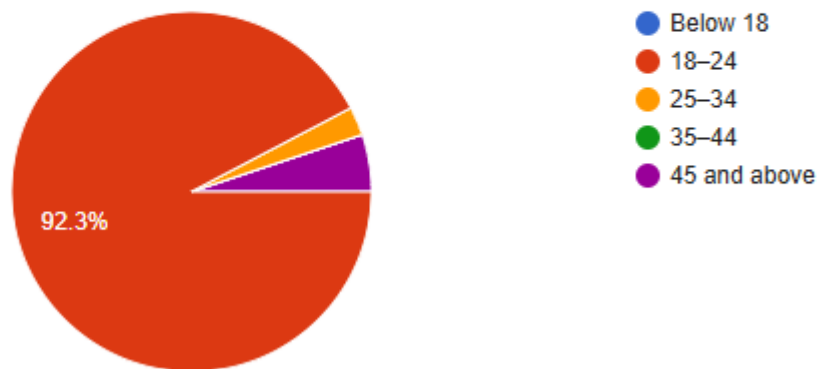
- Users have internet-enabled devices.
- Doctors provide accurate professional information during registration.
- Patients provide accurate information while creating accounts.
- Users will follow platform guidelines and policies.

Survey Questionnaires & Results

Section 1: Demography

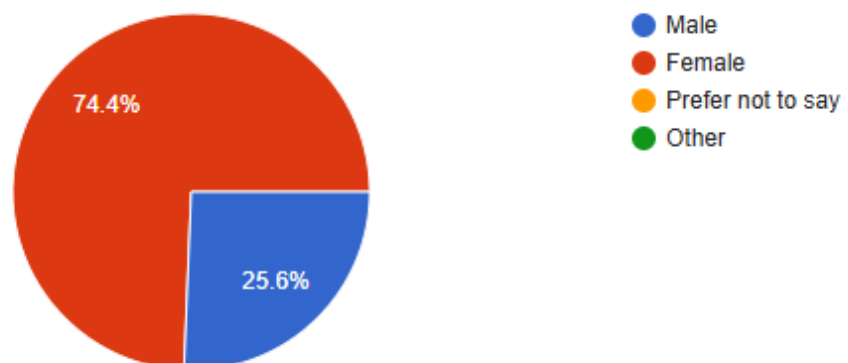
1. What is your age group?

39 responses



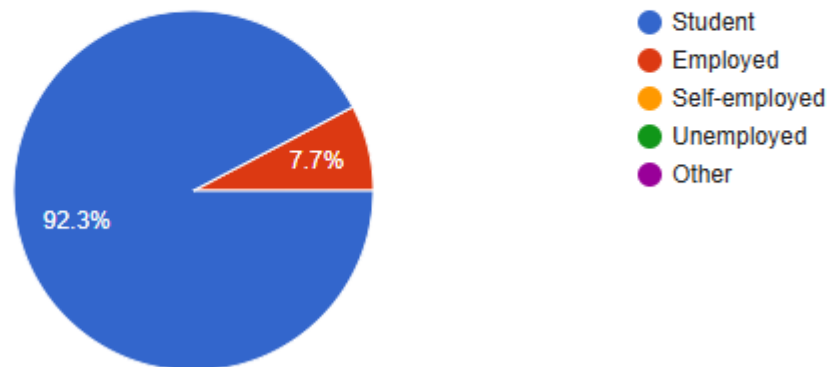
2. What is your gender?

39 responses



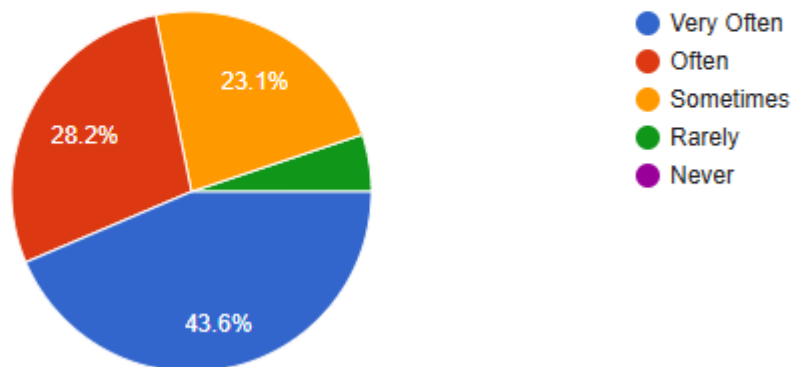
3. What is your occupation?

39 responses



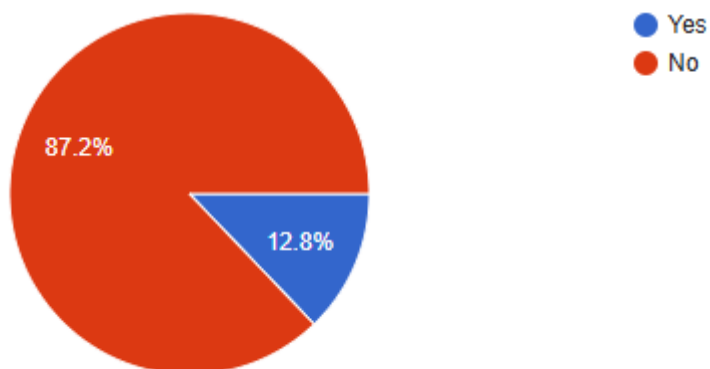
4. How often do you feel stressed due to academic, professional, or personal responsibilities?

39 responses



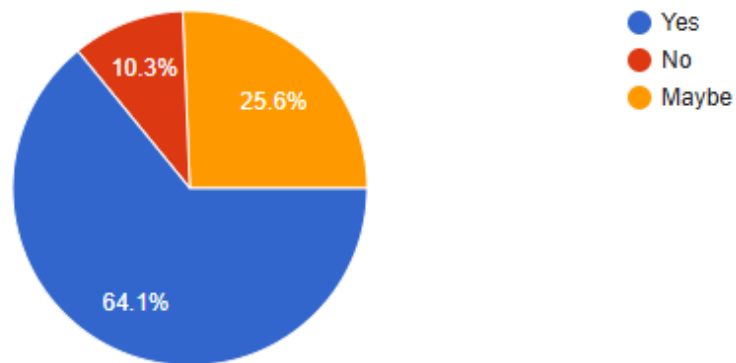
5. Have you ever used any online mental health or wellbeing service before?

39 responses



6. Would you be interested in using technology-based solutions to manage your mental wellbeing?

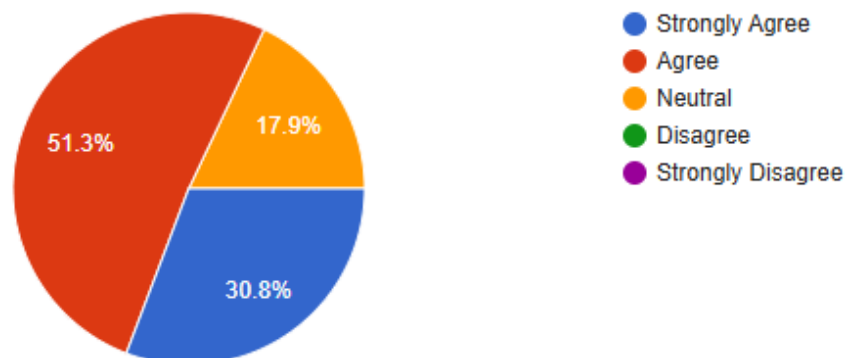
39 responses



Section 2: Functionality Assesment

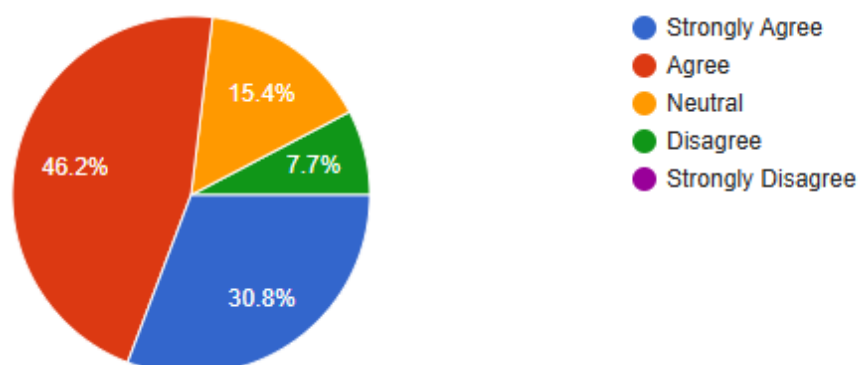
1. It is difficult to find access to mental health support when needed.

39 responses



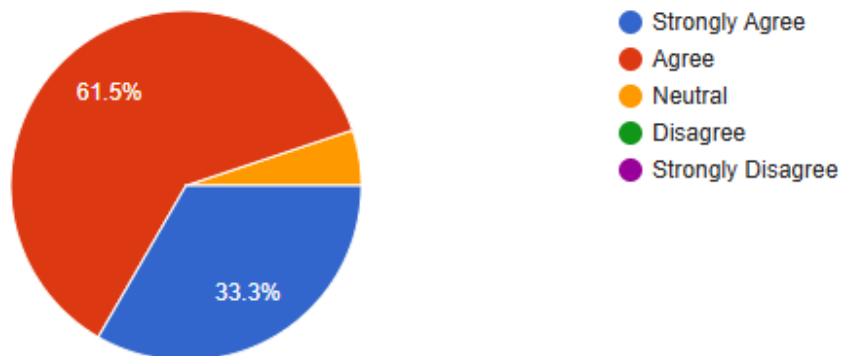
2. An online mental wellbeing management system would make mental health services more accessible.

39 responses



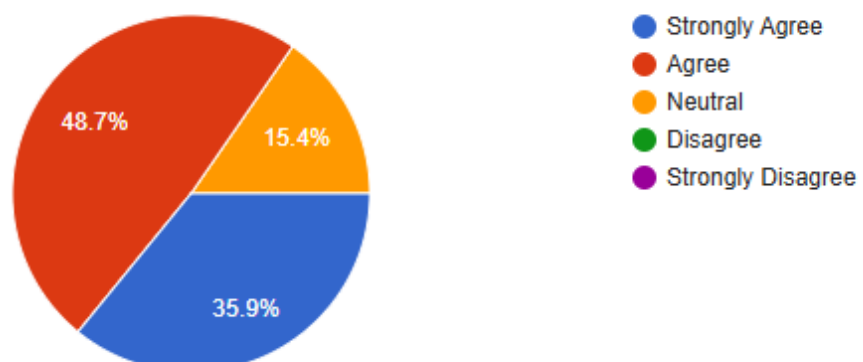
3. A centralized platform connecting patients, doctors, and administrators would improve mental healthcare services.

39 responses



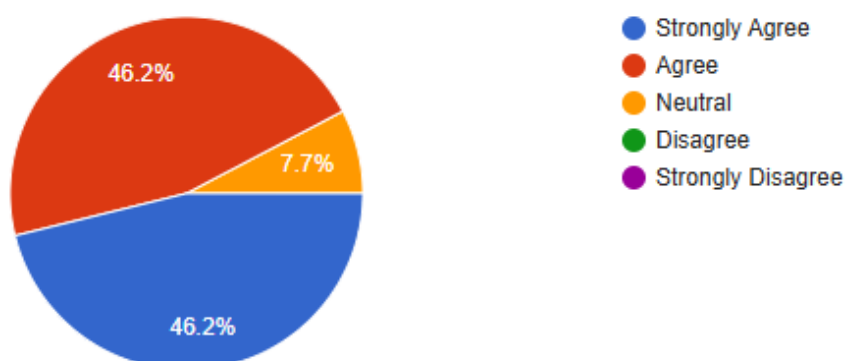
4. The ability to book appointments online would be useful for me.

39 responses



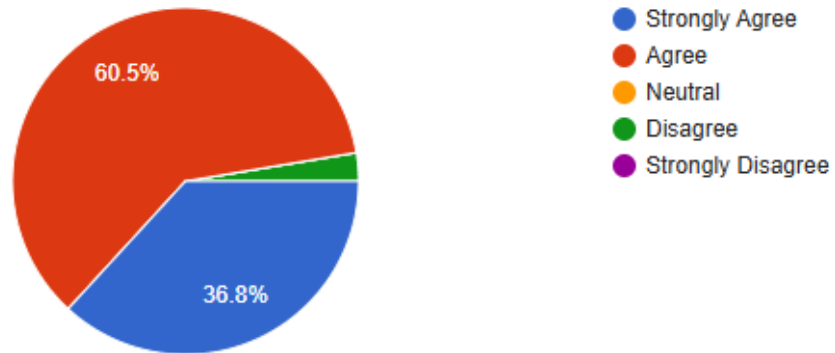
5. Viewing available doctors and their profiles before booking an appointment is important.

39 responses



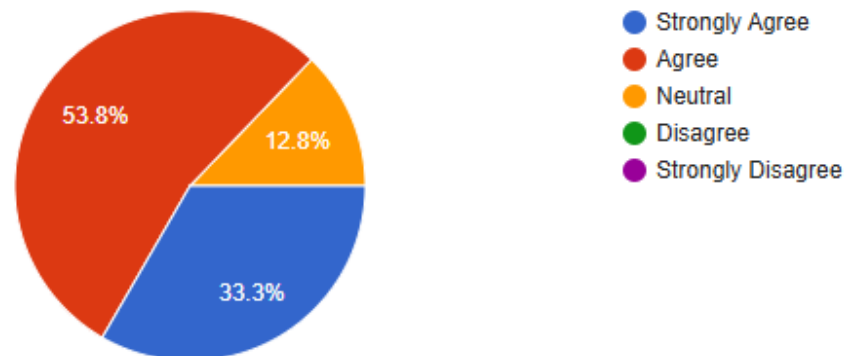
6. A mental health self-assessment feature would help users better understand their mental condition.

38 responses



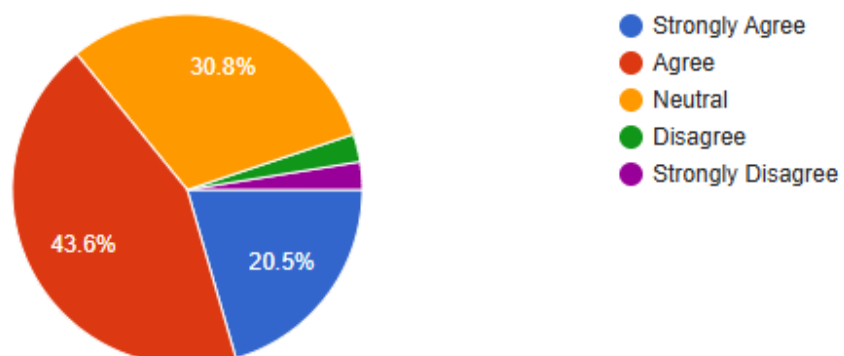
7. It will be very helpful to use self-assessment questionnaires before consulting a doctor.

39 responses



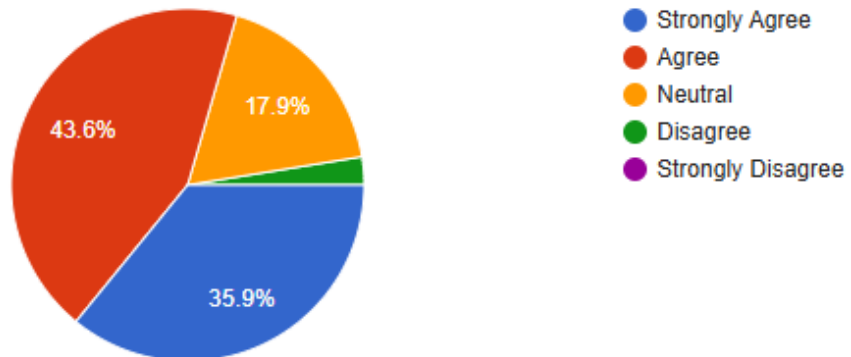
8. An AI chatbot can provide useful initial support and guidance for mental wellbeing.

39 responses



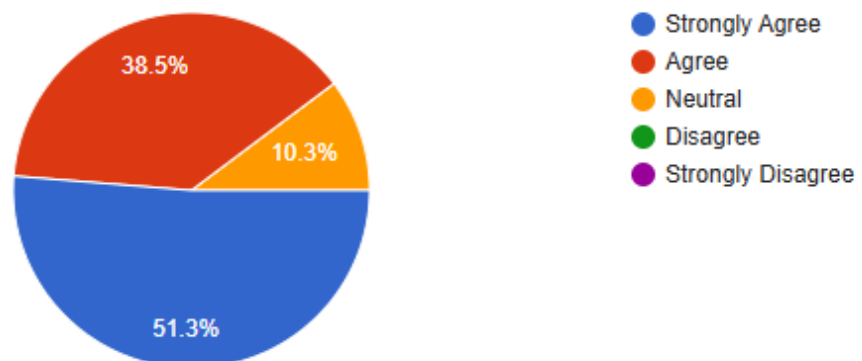
9. Having access to appointment history would improve the user experience.

39 responses



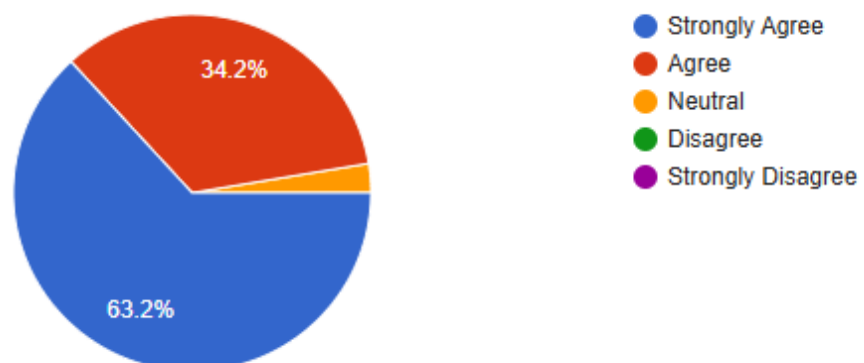
10. The doctor enrollment and verification process is important for ensuring service quality and trust.

39 responses



11. Protecting users' mental health information and privacy is very important.

38 responses



Do you have any suggestions or comments about the features or designs of this system? Feel free to share your thoughts.

12 responses

No

Very good initiative. 🍌

If you want to work this, so please do it but not just like a presentation.

It would also be helpful to include mental health resources, articles, and progress-tracking features for users.

No suggestions.

The system is user-friendly, but adding advanced search and real-time notifications could improve its functionality.

The system has a clean design and works efficiently for managing library operations.

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